EKURHULENI UNDERGROUND

The hidden systems of the East Rand

THEME ANALYSIS — ISSUE 8

The Springs Corridor

Why this theme matters for the long-term economic story of Ekurhuleni

The theme of the Springs Corridor is not a story about decline. It is a story about metamorphosis — about what happens when an economy built on underground extraction runs out of ground to extract, and what grows in the archaeology it leaves behind. The gold reef that made the East Rand famous, that drove the original urbanisation of Springs, Brakpan, Nigel, and Kwa-Thema, is largely exhausted. But the infrastructure, the labour traditions, the land patterns, and the chemical knowledge that the mining era deposited are not gone. They are being quietly repurposed.

The deep story this week: how the derelict mining belt running along the R50 between Springs and Nigel is becoming the unlikely seedbed of a new industrial economy — one built around metals reclamation, artisanal processing, agricultural chemicals, and the informal trade networks that now thread through the townships the gold boom once built.

Three supporting micro-stories: the informal steel fabricators of Brakpan repurposing retired mine machinery into agricultural equipment for southern African markets; the used-parts dealers of Kwa-Thema who have quietly built a cross-border trade in reclaimed mining and automotive components; and the small chemical processors near Nigel whose operations serve the agricultural supply chain of Mpumalanga and Limpopo, almost entirely off the formal economy's radar.

The overlooked place: the R50 mine dump corridor between Springs and Nigel — a chain of orange-gold hillocks visible from the highway, which most drivers treat as scenery and which an informal economy is methodically dismantling, gram by gram.

The interesting statistic: the East Rand is estimated to hold over six billion tonnes of gold tailings across its legacy mine dumps — one of the largest concentrations of reclaimable mineral material anywhere on earth. The gold rush never truly ended. It simply went slower and smaller and informal.

WEEKLY ISSUE — 13 MARCH 2026

The Afterlife of the Gold Rush

What the old mining belt between Springs and Nigel is quietly becoming — and why its most important actors have never held a mining licence.

1. EDITORIAL

The Colour of Old Money

There is a particular shade of orange that belongs entirely to the East Rand. You see it first from the highway — mounded against the sky south of Springs, piled along the R50 toward Nigel, catching the early morning light in a way that looks almost cinematic, almost intentional, as though someone arranged it there. These are the mine dumps. Compressed history, physically. Hundreds of millions of tonnes of rock ground fine and stacked above the veld, the residue of a century of gold extraction. If you did not know what they were, you might think they were simply hills. But this part of Gauteng has no natural hills. Everything you see above the horizon was put there by people.

I have been driving past those dumps for years without stopping. Most people do. They are background, furniture, the unremarked scenery of the East Rand commute. And yet something about them has been pulling at my attention lately — not because of what they are, but because of what is happening to them. They are getting smaller. Slowly, quietly, almost imperceptibly. Someone is taking them apart.

This is not a metaphor. Across the Springs-Nigel corridor, artisanal and small-scale operators have been systematically processing tailings from legacy mine dumps for years, extracting residual gold, titanium minerals, and other recoverable metals using methods that range from the barely legal to the entirely informal. They work at the margins of a system that the formal mining industry has largely abandoned, operating on land whose ownership is often contested and whose regulatory status is genuinely ambiguous. They are, in a real sense, completing the work the original mining companies left unfinished.

But this is only the most visible layer of what is happening in the Springs corridor. Underneath it — or alongside it, more accurately — a broader economic reinvention is underway. The corridor's legacy assets are not just mineral. They are spatial: the wide industrial plots, the railway sidings, the chemical infrastructure, the population of technically skilled workers who spent their careers underground and now need somewhere else to apply what they know. These assets are being combined, slowly and largely without coordination, into something new. Not a replacement for the gold economy. Something more interesting than that: a successor.

This issue is an attempt to describe what that successor looks like — and to ask whether anyone is paying enough attention to help it grow.

2. THE LONG ESSAY

The Quiet Industrial Succession of the Springs Belt

Section I — The Observation

Drive south from Springs on the R42 toward Nigel and the road takes you through a landscape that economic geographers sometimes call 'stranded assets territory' — a polite term for the physical remains of an economy that moved on without a forwarding address. On your left, a headgear tower stands against the sky, its winding wheels frozen in the position they occupied the day the mine closed. On your right, a processing plant whose corrugated walls have been painted and repainted, now hosting a light engineering firm that makes agricultural water tanks. A little further, a row of mine workers' cottages that have been absorbed into the informal housing fabric of the surrounding township, their original geometry still faintly legible beneath the extensions and additions.

What is striking about this landscape is not its dereliction — much of it is, in fact, actively used and economically alive. What is striking is its density of accumulated industrial knowledge. The Springs corridor may be post-mining, but it is not post-industrial. The skills, the infrastructure, the physical plant, the social networks — all of it is still here, in various states of adaptation. What has changed is not the capacity for industry. What has changed is the raw material.

The gold rush never ended. It simply went slower, smaller, and informal — and in doing so, it became something more durable.

Section II — The Hidden System

To understand the Springs corridor's hidden economy, you have to start with the mine dumps themselves — but not in the way most analysts do. The conventional story is about large-scale tailings retreatment: listed companies, environmental impact assessments, mechanised processing plants that can handle millions of tonnes. That story is real, and companies like DRDGold operate significant retreatment operations in the region. But it is not the whole story, and arguably not the most interesting part.

Below the formal retreatment sector is a stratum of artisanal and semi-formal operators who process tailings material by hand or with small mechanical equipment. They are not licensed. They are not registered. They are, in the language of the Minerals and Petroleum Resources Development Act, zama zamas when they operate without authorisation — a term that carries significant stigma but describes, in many cases, rational economic actors filling a gap the formal system cannot. These operators know the mineral content of specific dump sections with a precision that no geological survey has documented. They have developed processing techniques, mostly hydrometallurgical, that extract value from material the original mines considered exhausted. Their margins are thin and their working conditions are often dangerous. Their economic output, if it could be measured, would surprise people.

But the dumps are only part of the picture. The broader hidden system of the Springs corridor operates through three intersecting networks. The first is the metals reclamation economy — the scrap dealers, the secondhand machinery traders, the informal foundries that smelt recovered metal into usable stock. Springs itself has one of the densest concentrations of scrap metal dealers in Gauteng, a fact almost entirely invisible in the city's official economic profile. The second network is the repurposed-skills economy: the ex-miners, ex-metallurgists, and ex-plant operators who have converted their technical knowledge into informal service businesses — assaying, small-scale smelting, equipment repair, and the fabrication of bespoke mining and agricultural equipment from reclaimed steel. The third is the cross-border components trade, centred in and around Kwa-Thema, where used mining and automotive parts are collected, sorted, and sold into the supply chains of smaller mining and agricultural operations across Zimbabwe, Zambia, and Mozambique.

These three networks are not formally connected. There is no Springs Corridor Informal Industrial Alliance. But they share infrastructure, labour, and knowledge in ways that function like a cluster — in the sense that Michael Porter meant when he described the competitive advantage of geographically concentrated industries. They have, without intending it, created an informal industrial district.

Section III — The Broader Insight

The Springs corridor offers Ekurhuleni — and South Africa's urban planners — something genuinely rare: a legible case study in post-extractive economic transition. Most of the world's mining towns that exhaust their resource base simply depopulate. The Springs belt has not done this, and the reason is worth examining carefully.

Part of the answer is geography. The corridor sits within the gravitational field of the Gauteng megalopolis, which means it has retained market access and infrastructure connectivity that truly remote mining towns lack. Part of the answer is the scale of the original investment: the mining era built not just mines but entire urban systems — housing, roads, rail, water, power — that remain operational and useful. And part of the answer is the adaptability of the human capital the mining era trained. A man who spent twenty years operating underground rock-drilling equipment has a form of mechanical intelligence that transfers — imperfectly, creatively — to a wide range of industrial contexts.

What the Springs corridor suggests is that post-extractive economies do not need to be managed into new industries from above. They need to be observed carefully and supported at the margins:

given better infrastructure, better market access, and lighter regulatory touch in areas where the informal economy is already performing valuable functions. The challenge is institutional, not economic. The economic capacity is already there, expressed in a hundred small operations that nobody has thought to map.

Section IV — The Antifragile Angle

The informal industrial economy of the Springs corridor has a property that should interest anyone thinking seriously about economic resilience: it is almost impossible to damage from the outside. The formal mining sector has been hit, in the past twenty years, by commodity price crashes, energy crises, labour disputes, and regulatory upheaval. Each of these shocks reduced employment in the large mines and increased the flow of skilled workers and reclaimed equipment into the informal economy. The crises that hurt the formal sector fed the informal one.

This is antifragility in its most concrete form. When Anglo American restructures, former employees take their metallurgical knowledge and open informal assay services. When a large scrap dealer closes a regional depot, the network of small dealers absorbs its customer base within weeks. When formal retreatment companies suspend operations for environmental compliance reviews, the artisanal processors move into the gap — not noisily, not visibly, but persistently.

The policy implication is counterintuitive: the Springs corridor's informal economy is not a problem to be solved through formalisation. It is a system to be respected and carefully extended. The question is not how to replace informal operators with formal ones, but how to give informal operators access to the things — stable land tenure, basic financial services, digital market access — that would let them grow without destroying the flexibility that makes them resilient. That is a different kind of economic development policy than South Africa has traditionally attempted. It is also, arguably, the one that the evidence suggests would work.

3. THE HIDDEN ECONOMY

Three Operations Nobody Is Watching

1. The Mine-Steel Fabricators

BRAKPAN INDUSTRIAL ZONE

In a cluster of corrugated-iron workshops along the eastern edge of Brakpan's industrial strip, a small network of fabricators has built a niche business repurposing retired mine shaft components — winding drums, compressed-air receivers, structural steelwork — into agricultural equipment: grain storage silos, water-tank frames, irrigation infrastructure. The raw material is essentially free: end-of-life mining hardware that the formal scrap market prices at melt value. The fabricators' labour and engineering knowledge convert it, at relatively low cost, into products that small-scale farmers in Zimbabwe, Zambia, and Mozambique are willing to pay a meaningful premium for. The supply chain is invisible from the front; the demand comes through informal trade networks built on years of cross-border relationship-building.

2. The Kwa-Thema Parts Corridor

KWA-THEMA / SPRINGS

Along the main arteries running into Kwa-Thema, a dense concentration of used-parts dealers has evolved into something approaching a regional clearinghouse for reclaimed industrial and automotive components. What began as a township repair economy — the familiar network of backyard mechanics and roadside sellers — has quietly extended into a regional export operation. Buyers from mining operations in Zimbabwe, agricultural repair workshops in Mozambique, and small-fleet transport operators across southern Africa make regular buying trips to the Kwa-Thema corridor, sourcing parts that are unavailable or unaffordably priced through formal

channels. The dealers know their inventory with encyclopaedic precision. They do not have catalogues. They have memory.

3. The Nigel Agrochem Processors

NIGEL / HEIDELBERG ROAD CORRIDOR

South of Nigel, in the corridor toward Heidelberg, a handful of small chemical processing operations have established themselves in the margins of the former mining chemical supply chain. They produce, or more accurately blend and reformulate, crop protection and soil treatment products — fungicides, micronutrient solutions, pH adjusters — using equipment and chemical knowledge inherited from the mining reagents industry. Their customers are small commercial farmers in Mpumalanga, Limpopo, and the North West, who buy on price and on the relationships they have built with the operators. These are not registered agrochemical manufacturers in the formal regulatory sense. They occupy an ambiguous space that the formal industry prefers not to look at directly. Their products work. Their margins depend on it.

4. THE MAP OF THE WEEK

The R50 Dump Belt, Springs to Nigel

26°14′S 28°27′E → 26°22′S 28°27′E

Approximately twenty-two kilometres. That is the length of the R50 corridor running south from Springs to Nigel along which the mine dump landscape is most concentrated and most active. In those twenty-two kilometres you will find: the large-scale DRDGold retreatment operations visible from the highway; the smaller, less visible artisanal processing sites at the dump perimeters; the scrap metal dealers who have colonised the old mine access roads; and the fabrication workshops that have grown up in the service zones behind the dumps.

What you will not find is a map. No tourism board has charted this corridor, no heritage marker explains the geology, no economic development report has totalled the turnover. The orange hillocks rise and diminish on their own schedule, processed by hands that no economist has counted. To drive the R50 at dusk, when the dumps catch the last light and the dust from the day's processing hangs in the air above the loading bays, is to understand that the East Rand is not finished with its history. It is simply working through it at a different speed.

5. THE EAST RAND NUMBER

6 billion tonnes

Estimated gold tailings material in East Rand legacy mine dumps

The Witwatersrand Basin — of which the East Rand is a central component — is estimated to contain upwards of six billion tonnes of gold tailings in its legacy mine dumps. At current gold prices and using modern retreatment technology, a significant portion of this material is economically recoverable. The formal retreatment sector processes millions of tonnes annually. The informal sector processes an unknown additional quantity. The gold rush, which began in 1886, has not ended. It has simply changed characters: from hard-rock mining companies to retreatment specialists to artisanal operators who work the same ground by different methods. The veld remembers everything.

6. CLOSING REFLECTION

What the Dumps Leave Behind

There is a version of the Springs corridor's future that is easy to dismiss because it looks, at first glance, like a past. The orange hillocks, the frozen headgears, the repurposed industrial sheds — all of it carries the texture of an economy in decline. But look again, and the movement is there: the trucks taking mine steel to the fabricators in Brakpan, the buyers from Harare moving through the parts dealers in Kwa-Thema, the agricultural traders pulling up to the Nigel processing yards with empty containers and leaving with full ones.

The Springs corridor is not waiting for investment to arrive and tell it what to become. It is already becoming something — a post-extractive industrial district built on reclaimed materials, reclaimed skills, and reclaimed knowledge, connected to markets across the subcontinent by networks that owe nothing to formal economic planning. The question is not whether this economy is real. It is whether anyone with the power to support it is paying close enough attention to do so wisely, before the impulse to formalise destroys the very flexibility that makes it work.

— Ekurhuleni Underground, 13 March 2026

Next issue: The Alberton-Germiston Manufacturing Ridge — the quiet persistence of heavy industry in the shadow of a service economy.